

MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY JAIPUR
Department of Physics

Mid-term Examinations 2026

B. Tech: Even Semester
Date: February 25, 2026

Course: Modern Physics
Duration: 1.5 Hours

Course Code: 22PHT102
Max Marks: 30

Answer all the questions.

- Q1. (a) Evaluate $\oint_C [(2x - y)dx - yz^2dy - y^2zdz]$, where c is the circle $x^2 + y^2 = 1$, corresponding to the surface of a sphere of unit radius. [4]
- (b) Write the physical significance of the divergence and curl. [3]
- (c) If $E = [(2y^2 + z)\hat{i} + 4xy\hat{j} + x\hat{k}] V/m$ and $\epsilon_0 = 8.854 \times 10^{-12} F/m$, find out the volume charge density at $(-1, 0, 3)$. [3]
- Q2. (a) What is the significance of displacement current density? [2]
- (b) Derive the continuity equation in electrodynamics and discuss its physical significance. [4]
- (c) Write Maxwell's equations in differential and integral forms. [4]
- Q3. (a) Derive the wave equation for electromagnetic waves in free space. [4]
- (b) A uniform plane electromagnetic wave propagates in free space and has an electric field given by $\vec{E} = \hat{j} 10 \cos(10^8 t + 20z) V/m$, Determine the direction of propagation of the wave, polarization of the wave, magnetic field $\vec{B}$ and the average energy carried by the wave. [1 + 1 + 1.5 + 2.5]

$\frac{10 \times 3}{10^{-2}} = 3 \times 10^8$

$\vec{E} = \hat{j} 10 \cos(10^8 t + 20z) V/m$

$\vec{B} = \frac{1}{c} \frac{\partial \vec{E}}{\partial t}$

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$\vec{B} = \frac{1}{3 \times 10^8} \times 10 \times 10^8 \sin(10^8 t + 20z) \hat{k}$

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$\vec{\nabla} \times (\vec{\nabla} \times \vec{E}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \nabla^2 \vec{E}$

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Malaviya National Institute of Technology Jaipur
Department of Physics
Midterm examination 2024-25

Class: B.Tech. 1st semester.
Course Title: Classical Physics.
Course Code: 22PHT101.

Date: September 24, 2024.
Time duration: 1 hour 30 minutes.
Maximum Marks: 30.

- Q.1) (a) A particle moves along the curve $r = r_0 e^{\alpha\theta}$ with $\dot{\theta} = \omega$ constant, where r_0 and α are constants. Find the velocity and acceleration of the particle in polar coordinates and find the value of α for which the radial part of the acceleration is zero. 4 marks
- (b) Explain the inertial and non-inertial frame of reference with an example. 3 marks
- Q.2) (a) Find the center of mass of a homogeneous right-angled triangular plate of negligible thickness. The perpendicular sides both measure to length L and the total mass is M . 4 marks
- (b) A rocket is lifted-off the surface of the earth with no horizontal motion at all. The exhaust velocity of the fuel expelled by the rocket is $\vec{u}$ (constant) relative to the rocket. The mass and the velocity of the rocket is M and $\vec{v}$ at time t respectively. Obtain the equation of motion of the rocket. If the initial velocity of the rocket at time $t = 0$ is zero and the initial mass is M_0 , find the final velocity in terms of the final mass at time $t = t_f$ assuming a constant gravitational field. 5 marks
- Q.3) A particle is under the influence of a one dimensional force

$$F = -kx + \frac{kx^3}{\alpha^2},$$

where k and α are constant, and k is positive. (a) Find the potential energy at any point x by assuming that the potential energy is zero at $x = 0$. (b) At which location the potential energy is $\frac{1}{4}k\alpha^2$. (c) Which are the other points where the potential energy for this system is zero, apart from the origin. 5 marks

Q.4) Derive the equation of motion for a harmonic oscillator experiencing a damping force of magnitude bv , where b is constant and v is the velocity of the particle. Solve it to find the position as a function of time for the under-damping case. What do you mean by critical damping and over-damping. 9 marks

Malaviya National Institute of Technology Jaipur

Department of Physics
Mid Term Examination 2024-25

B.Tech. 2nd Semester

Course: Modern Physics (22PHT102)

Date: February 24, 2025

Duration: 1:30 hours

Max. Marks: 30

Answer all the questions

1. The x and z components of a static magnetic field in a region are $B_x = B_0(x^2 - y^2)$ and $B_z = 0$, respectively. Find out the y component using the Maxwell's equation. (2)

2. Prove that the Ampere's law is not applicable for non-steady currents. (2)

3. The height of certain hill in meter is given by
$$h(x, y) = 10(2xy - 3x^2 - 4y^2 - 18x + 28y + 12)$$
where x and y are respectively the distances (in km) towards east and north of a reference point named 'A'. Find the location of the top of the hill with respect to reference point A. (3)

4. Calculate the line integral of $\vec{v} = (2xz + 3y^2)\hat{y} + (4yz^2)\hat{z}$, over a square of unit dimensions. (3)

5. Prove that the magnetic force does not speed up or slow down a charged particle. (3)

6. The electric field of an electromagnetic wave is propagating in a vacuum represented by $E(y, t) = E_0 e^{i(ky - \omega t)} \hat{z}$ V/m. If the amplitude of the electric field of the wave is 5 V/m and the wavelength is 0.10 m. Then calculate the average rate at which energy is transported by the wave (per square meter). (3)

7. Derive and explain the continuity equation in electrostatics. (3)

8. Derive the electromagnetic (EM) wave equation and determine the speed of the EM waves. (3)

9. Write and explain the Maxwell equations in integral form (corrected form). Starting from the integral form, deduce these equations in differential form. (8)

2023 VAI III

Malaviya National Institute of Technology Jaipur

Department of Physics

B.Tech II Semester Mid Term Examination 2023-24

Course: Modern Physics

Course Code: 22PHIT102

Date: February 27, 2024

Duration: 1 hour 30 minutes

Max. Marks: 30

Answer all the questions

1. (a) State and explain the significance of de Broglie's hypothesis. (1+2)
(b) Distinguish between phase velocity and group velocity. (1)
2. Write one dimensional time dependent Schrodinger's wave equation and from there derive the time independent Schrodinger's wave equation in one dimension. (1+4)
3. (a) Draw the wavefunctions and probability functions of ground state, first and second excited states, for a particle in 1-D infinite potential box of width 'a'. $n=1$, $n=2$, $n=3$ (3)
(b) For the case $n = 3$, find the probability that the particle will be located in the region $\frac{a}{3} < x < \frac{2a}{3}$. (2)
4. (a) Calculate the volume of the reciprocal lattice of the face centered cubic lattice. (3)
(b) A beam of X-rays of wavelength 0.071 nm is diffracted by (110) plane of rock salt (FCC) with the glancing angle of 21° for the second-order diffraction. Find the lattice constant. (3)
5. (a) Find the relaxation time of conduction electrons in a metal of resistivity $1.54 \times 10^{-8} \Omega m$, if the metal has 5.8×10^{28} conduction electrons/ m^3 . (2)
(b) What is Fermi energy and Fermi surface? Derive the expression of electronic density of states (DOS) for a metal in 3-D. (2+3)
6. Write down Fermi-Dirac distribution function for the electrons (Fermions) and plot its characteristic variation with energy for temperatures $T = 0 K$, and $T \neq 0 K$. (1+2)

(Boltzmann constant: $1.38 \times 10^{-23} J K^{-1}$; mass of electron $9.1 \times 10^{-31} kg$; $h = 6.62 \times 10^{-34} Js$)